

2 Carnot cycles and Entropy

The problems are roughly ordered to coincide with the lectures. Those indicated with ♥ are especially important, i.e. 1.1 and 1.6. Any labelled with ♣ are the hardest ones, which might only occur as a "sting in the tail" at the end of a long examination question.

Problem 2.1 A Carnot cycle

A Carnot engine takes 2000J of heat from a reservoir at 500K, does some work and discards some heat to a second reservoir at 350K. Calculate:

- (i) The heat energy discarded to the cold reservoir per cycle [1400J]
- (ii) The work done per cycle [600J]
- (iii) The efficiency of the engine [30%]
- (iv) The entropy changes to the substance undergoing the cycle on each of the adiabatic and isothermal changes.

Problem 2.2 Second law – equivalence of Kelvin and Clausius

Show the converse of the result established in the lecture notes: that Kelvin's statement of the second law is implied by Clausius' statement of the second law.

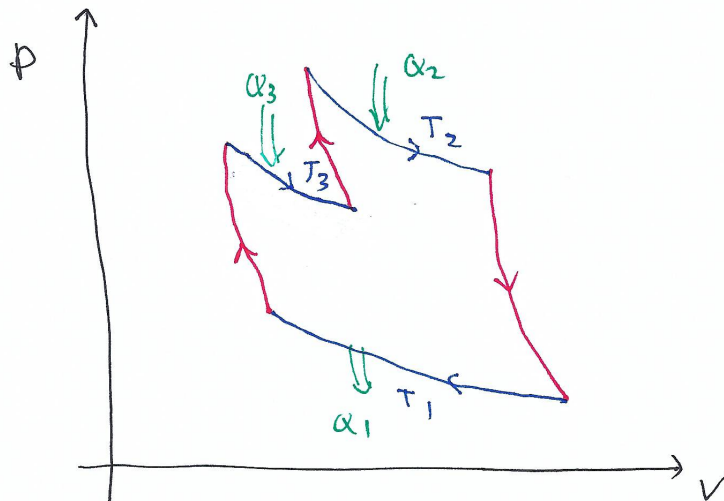


Figure 2.1: A cycle involving a third reservoir, T_3 .

♣ Problem 2.3 Three reservoirs do not improve efficiency

By considering the path in the p - V plane with a third isotherm, see Fig. (2.1), with $T_2 > T_3 > T_1$, show that this cycle is less efficient than that only involving two reservoirs, T_2 and T_1 , if one extends the T_2 isotherm to the left to intersect an extended left adiabat.

Problem 2.4 Integrating Entropy

For a small reversible change carried out to a system, the first law can be written in the form $dU = TdS - pdV$. Use this equation to derive the expression for the entropy of one mole of an ideal gas:

$$S(V, T) = S(V_0, T_0) + C_v \ln(T/T_0) + R \ln(V/V_0)$$

using a judiciously chosen path between (V_0, T_0) and (V, T) in the V - T plane.

Problem 2.5 Homely entropy examples

What entropy changes occur in the universe when:

- (i) A 100 W light bulb is left on for 1 minute in a large room at 20C [20.5JK⁻¹].
- (ii) 0.2 litre of water at 80C is thoroughly mixed with 0.8 litre of water at 10C [17JK⁻¹]

You should assume that these systems are isolated from their surroundings, so there is no need to consider entropy changes to the national grid in part (i) or to whatever mixes the water in part (ii). Take the specific heat of water to be 4190 J kg⁻¹K⁻¹.

Problem 2.6 Water to steam

Use the information given below to calculate the difference in internal energy between 1 kg of steam at atmospheric pressure and 100 C and the same quantity of water at 0 C [2511kJ].

Specific heat capacity of water	= 4.2 kJ kg ⁻¹ K ⁻¹
Latent heat of evaporation at 100 C (at constant pressure)	= 2.26 × 10 ³ kJ kg ⁻¹
Volume at atmospheric pressure – water	= 1.02 × 10 ⁻³ m ³ kg ⁻¹
Volume at atmospheric pressure – steam	= 1.67 m ³ kg ⁻¹
Atmospheric pressure	= 1.01 × 10 ⁵ Pa

Problem 2.7 Entropy of a cooling block

A block of heat capacity $C = 500 \text{ JK}^{-1}$ at $T_1 = 80^\circ\text{C}$ is cooled down by contact with a reservoir at temperature $T_2 = 10^\circ\text{C}$. What is the change of entropy of the block in the process? Of the reservoir? [-110 JK⁻¹, 124 JK⁻¹]

Problem 2.8 Joule's second experiment

A closed cylinder of volume 0.4m³ is separated into two halves by a thin strong membrane that holds an ideal gas at 2 bar and 25C on the one side and is empty on the other. The membrane is then pierced so that the gas squirts into the empty side. Calculate the entropy change of the gas.

[Hint: remember that entropy is a function of state, so the problem can also be tackled by considering what would happen if the gas were reversibly recompressed to its original volume.] [93 JK⁻¹]

Problem 2.9 Maximum work out of a hot block

Consider a block of metal with heat capacity $C = 500 \text{ JK}^{-1}$ and original temperature $T_2 = 200^\circ\text{C}$. The block is used as the hot source to drive a heat engine in contact with a cold reservoir at $T_1 = 20^\circ\text{C}$. What is the final temperature of the block? What is the maximum work that can be extracted from the engine? What efficiency does that represent? [Hint: you can assume that the temperature of the block does not significantly change over a single cycle of the engine.] [20 kJ, 0.22]

Problem 2.10 T - S diagram

Draw a Carnot cycle (for an engine) in a T - S diagram. Calculate heat received by the system, and deduce a geometric interpretation of the work done by the engine. Can this be generalised to an arbitrary cycle drawn in a T - S diagram, where T is the applied (external) temperature?

Problem 2.11 Entropy of a p - T change

One mole of ideal gas with specific heat capacity $c_p = \frac{5}{2}R$ undergoes a transformation from its initial state ($T_1 = 20^\circ\text{C}$, $p_1 = 1 \text{ atm}$) to a final state ($T_2 = 80^\circ\text{C}$, $p_2 = 1.5 \text{ atm}$). Find the entropy change of the gas. [3.9 JK^{-1} - 3.4 JK^{-1}]

♣ Problem 2.12 Two-phase cycle

A Carnot cycle comprises a specified sequence of reversible isothermal and adiabatic steps, but the working substance does not have to be an ideal gas. Suppose that we run a Carnot cycle between temperatures T_2 and T_1 using the usual gas-in-cylinder configuration, but such that the cylinder contains a small amount of liquid, and that the rest of the volume is filled with saturated vapour. The pressure of the saturated vapour depends only on the temperature.

- (i) What will be the efficiency of this engine?
- (ii) Sketch the p - V diagram for the gas.
- (iii) Consider the case where the change in temperature between the two isothermals, is small, such that $T_2 - T_1 = dT$ and $T_2 \simeq T_1 = T$. The adiabatic expansion is also very small, such that the pressure drops from p to $p - dp$. The increase in volume in the isothermal stage of expansion at temperature is ΔV and the latent heat required to evaporate the required amount of vapour is L . Evaluate the work done in going round a full cycle and hence obtain an expression for the efficiency in terms of ΔV , dp and L . By comparing this efficiency with the result from part (i), show that:

$$\frac{dp}{dT} = \frac{L}{T\Delta V}.$$

[You have now proved, by an unusual route, the well known Clapeyron equation for the temperature dependence of the vapour pressure of a liquid]